



SOIL TECH

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GEOTECHNICAL INVESTIGATION REPORT

PROJECT: Geotechnical investigation for the proposed G+3 Residential Building

LOCATION: # 37, Paramount Gardens, Talagattapura, Kanakpuraroad, Bangalore, Karnataka 560032

TOTAL NUMBER OF BOREHOLES: 3

CLIENT: Mr Rajkumar Hegade, Bangalore

AUGUST-2026

Report No. SI/GR/027/2026

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Report on :

Geotechnical Investigation for the
proposed construction of G+3 Residential
building

#37, Paramount Gardens, Talagattapura,
Kanakpura road, Bangalore, Karnataka
560109

Report for :

Mr. Rajkumar Hegade

Reference :

Mr. Divakar S
Structural Engineer
Mobile: 9901443608

**Laboratory investigation
and recommendations by :**

Dr. LAKSHMI V
Consulting Geotechnical Engineer

**Date of submission of
report:**

01.09.2026

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(I)BORE LOGS AND LABORATORY TEST RESULTS

(II)APPENDIX SHOWING TYPICAL SBC CALCULATIONS OF SOIL FOR ISOLATED FOOTING

(III) LOCATION PLAN OF BOREHOLES

Geotechnical investigation for the proposed construction of G+3 Residential building, At Talagattapura, Bangalore 560109

1. Introduction

Mr. Rajkumar Hegade., proposed for the construction of above project. Geotechnical Investigation was assigned to Mrs. Lakshmi V, Bangalore in view to furnish the detailed technical information of the nature of subsoil strata for detailed analysis and foundation designs.

The report covers the details of field investigation and the tests carried out in the laboratory with the recommendation of net SBC based on the soil samples collected by Manual Auguring.

2. Scope of Work

The geotechnical investigation at the proposed location has been carried out by manual auguring **3 bore holes namely BH-1, BH-2 and BH-3.**

Scope of geotechnical investigation includes the following:

- a) To identify the soil stratification up to a maximum depth of 6m or refusal strata.
- b) To conduct SPT test in boreholes generally at 1.5m and collection of representative samples.
- c) To identify the water table and seepage water if any
- d) To conduct the following laboratory test on representative samples:
 - Grain Size Analysis
 - Natural Moisture Content
 - Atterberg Limits
 - Shear Strength Parameters

Compilation of field and laboratory test data and submission of report.

3. Objectives and Scope of Work

a) Objectives

The primary objectives of the Geotechnical Investigation are:

- Determination of the nature of the deposits of soil
- Determination of the depth and thickness of the various soil strata and their extent in the horizontal direction

- The location of ground water and fluctuations in GWT
- Obtaining soil samples from the various strata
- Determination of the engineering properties of the soil strata that affect the performance of the structure, and
- Determination of the in-situ properties by performing field tests
- To recommend the type and depth of foundation
- To recommend improvements to the weak soil strata, if any.

b) Scope of work

The scope of work includes the following field (In-situ) and laboratory tests.

3.1 Field (In-situ) Investigations

- i. Boring 150mm/75mm dia boreholes in all kinds of soils strata using manual auguring.
- ii. Conducting field tests such as Standard Penetration Tests as per IS 2131- 1981(RA-2016) at various depths in the boreholes.
- iii. Collection of undisturbed soil samples at specified depths for determination of cohesion (C) and friction factor (Φ) as per IS-2132 - 1986.
- iv. To record the ground water table level if any.

3.2 Laboratory Testing Procedures

3.2.1 Physical and Engineering Properties of Soil

The scope of laboratory testing as per the relevant parts of IS: 2720

3.2.2 Report

This report comprises of soil profiles, recommendations regarding allowable bearing capacity, type and depth of foundations and improvement in bearing capacity if any & submission of report with relevant recommendations.

4 Schedule of Investigation

4.1 Method of Field Investigation

To study the subsoil strata, **3 nos.** of 150mm size dia boreholes were drilled below the existing ground level using manual auguring at the specified locations.

Table.1 Bore Hole Termination Depth

BORE HOLE LOCATION	DEPTH FROM EXISTING GROUND LEVEL (m)
BH-1	1.5
BH-2	4.0
BH-3	2.3

4.2 Sampling

From the Standard Penetration tests, disturbed samples have been collected by using split spoon sampler.

4.3 Standard Penetration Tests-IS2131

Standard Penetration Test (SPT) for each borehole location was conducted at every 1.5 m interval. SPT test utilizes a 50 mm OD and 600 mm long thick- walled split sampler tube driven into undisturbed soil under the impact of 63.5 kg weight hammer having a free fall of 750 mm. The penetration resistance of 'N' value is recorded as the number of hammer blows required to achieve a penetration of 300 mm after the initial penetration of 150 mm as a seating drive (to allow for any disturbed materials at the bottom of borehole). After completion of test, the sampler tube is removed and disassembled to provide a disturbed but representative sample. In all the boreholes the SPT was conducted up to refusal strata or until rock was encountered. Relationship between N Value and relative density/consistency of soil is presented in **Table 2**. **Figure 1** shows the Standard Penetration test at site.

**Table 2: Relative Density and Consistency of Soil Based on SPT
(Ref: Terzaghi and Peck, 1948)**

SAND		CLAY	
SPT N Value	Relative Density	SPT N Value	Consistency
0-4	Very loose	<2	Very soft
4-10	Loose	2-4	Soft
10-30	Medium	4-8	Medium
30-50	Dense	8-15	Stiff
>50	Very Dense	15-30	Very Stiff
		>30	Hard



Figure 1: Standard Penetration Test at site

5 General Geology and Note on Seismicity Features

The Soils in and around Bangalore comprise of predominantly red laterite / reddish brown fine loamy to sandy soils (with clay as interstitial binder) with a wide variation in overburden thickness. Below top Soil occurs weathered rock exhibiting varying degree of weathering with degree of weathering generally decreasing with depth. Further below, major part of Bangalore region is covered by Younger Gneiss complex representing remobilized parts of an older crust with abundant additions of newer granitic material. These Gneisses falling within the age group of 3.0 to 2.6 billion years are confined to the wide belt of gneisses surrounding the Achaean nucleus. This complex encloses narrow schist belts (Kolar type), largely basaltic in composition and characterized by gold mineralization. Such are the schist belts of east Karnataka like those of Kolar, Hutti, Maski and other belts. The gneissic rocks falling within the YGC are mainly granodioritic and granitic in composition. The bedrocks essentially consist of granites and gneisses intruded by number of basic dykes.

Seismicity

Bangalore city falls under Zone II of seismic map as per IS: 1893 (Part I) -2002. Recent earthquakes that occurred close to Bangalore were in the range of 2 to 5.5 in Richter scale. On January 29, 2001, earthquake magnitude of 4.3 in Richter scale hit in the Mandya area, with its epicentre about 35km south of Bangalore.

6 Seepage Water/ Ground Water Table

The ground water table/Seepage water was not encountered at the time of field investigation.

7 Sub Soil Profile

Location wise sub soil profile analysis and stratification. Stratification consists of **filled up** to a depth varying from **1m @ BH-1 and 0.5m @ BH- 2 & 3** from existing ground level.

The stratigraphy of those exploratory boreholes is summarized in **Table 3**.

Table 3: Stratigraphy of BH-1, 2 and 3

BH No	Depth (m)	Description/Stratigraphy
BH-1	0-1.5	Brownish Clayey Silty Sand with traces of Gravel
BH-2	0-1.5	Brownish Clayey Silty Sand with traces of gravel
	1.5-3.0	Brownish silty clay with traces of Soft Disintegrated Rock
	3.0-4.0	Soft Disintegrated Rock
BH-3	0-1.5	Brownish Clayey Silty Sand with traces of gravel
	1.5-2.3	Brownish Clayey silty sand with gravel

The 'N' values recorded at various depths in the borehole are indicated in respective bore log.

8 BEARING CAPACITY ANALYSIS

At depths ranging from 0 – 6.0m from existing ground level, depending on stratum encountered, in-situ compactness observed and net SBC calculations have been made based on the following criteria.

8.1 Net SBC based on shear failure criteria as recommended by IS:6403-1981

The calculations are based on "TERZAGHI" bearing capacity equation as recommended by IS: 6403 which takes care of L/B ratio (shape), foundation depth etc., along with other parameters. The intensity of loading at the base of the foundation, which would cause shear failure of the soil support, is termed as ultimate bearing capacity (UBC). The following equation is used to calculate net ultimate bearing capacity in semi-cohesive soils.

$$Net\ UBC = C*NC*Sc*dc + \gamma*D*(Nq-1) *Sq*dq + 0.5*\gamma*B*N\gamma*S\gamma*d\gamma$$

Net Safe bearing capacity (SBC) is the maximum intensity of loading that the foundation will safely carry without the risk of shear failure of soil irrespective of any settlement that may occur. Net Safe bearing capacity can be obtained by dividing the Net UBC with a factor of safety generally 3.0.

$$Net\ SBC = Net\ UBC/3.0$$

For Typical Safe bearing capacity calculations - Refer Appendix.

Net ABP based on Immediate Settlements

Soil being predominantly Sandy /Silty, and dry consolidation settlements is not expected to occur. Settlements, if any will be immediate. Hence as advised in IS: 8009 (Part I) code Para 9.1.4, the settlement of footing is estimated for a given pressure using N value recorded at the designated founding depth by referring to Fig No 9. Over and above, in standard Geo-technical practices, maximum allowable settlement is 25mm. Hence net allowable pressure is computed for this settlement.

Net Allowable Bearing pressure (ABP) is the intensity of the loading, which the foundation will carry without undergoing settlement in excess of the permissible value for the structure under consideration but not exceeding the net safe bearing capacity (SBC). Here, net allowable bearing pressure (ABP) is computed by referring 'N' values available from 2m and 3m depth as per IS-8009 (Part 1) on a conservative basis. As per IS-8009 (Part I), settlement for 1kg/cm² for various foundation widths varying depths available depending on design SPT 'N' values.

9 Conclusions

The following conclusions are given based on field and laboratory investigations.

1. The SPT values indicate the soil stratum consists of medium dense to dense formations.
2. The ground water table was not encountered during the time of investigation

3. The sub strata revealed filled up to a depth varying from 1m @ BH-1 and 0.5m @ BH-2 & BH-3 filling from existing ground level, any filled up layer as foundation base should be avoided as it may cause shear / settlement failure of foundation.
4. Laboratory test results indicate the soil is predominantly cohesionless in nature and as per grain size analysis the soil consists of predominantly sandy Silt with few traces of clay binder/sandy silt with clay & mica/ silty Sand.
5. It is recommended to adopt soil nailing /sheet piling /micro pile to protect the vertical sides of excavation if found any caving in.

10 Recommendations

The following recommendations are made based on the detailed investigation conducted and the conclusions drawn. The recommendations are restricted to the location around investigation points only.

10.1Type of foundation

The structure shall be founded on shallow foundation. This may be isolated footing or combined footing. It is advisable to go for a raft foundation if the footing area exceeds 50% of the plinth area or it is proposed to have a basement.

10.2Depth of foundation

It is suggested that Isolated / combined footing should be founded at a depth of 2.5/3.0m from existing ground level / Level of basement which would be adequate for the proposed type of structure based on design loads.

10.3 Safe bearing capacity of footings

Based on shear failure criteria, the safe bearing capacity and based on settlement criteria the allowable bearing pressure has been worked out and specimen calculations for the same is shown in appendix.

The recommended safe bearing capacity has been presented in the following **Table.4** for a conservative design purposes.

Table.No.4. The values of Safe bearing capacity at different depths at the proposed Bore-Hole location.

Bore Hole No.	Width of foundation (m)	Depth of foundation (m)	Safe bearing capacity (kN/m ²)	Safe bearing capacity (tons/m ²)
BH No.1	1.0	1.5	160	16
BH No.2	1.0	1.5	150	15
		3.0	240	24
		4.0	320	32
BH No.3	1.0	1.5	160	16
		2.3	220	22

11 Precautions

- Profile diagrams should be taken as indicative with regards to soil profile. The columns should be tied with R.C.C beam at plinth level.
- If any loose pockets of soil or filled up soil wherever encountered should be completely removed and back filled with well compacted earth. Thereafter a layer of 40-50mm size stone aggregates should be rammed into the back filled earth. A leveling course of lean concrete should then be laid over the aggregate course and construction of foundation can be taken up subsequently.
- The stability of foundation against overturning should be checked. While dimensioning the foundations the factor of safety against overturning should be as per IS:1904 (1986), art, 17.2

12 Special Note

- Whenever loose pockets of soil are encountered; it may be necessary to increase the depth of foundations.

- All depth of foundations is given with respect to top of existing ground level.

13 Remarks

1. The recommended safe bearing capacity values are less than computed to take into account rise of ground water table conditions, effect of eccentric loads if any, the complexity of soil fabric, texture and soil behavior.
2. The ϕ value is obtained for the corresponding SPT value 'N' as per the Fig.1 as given in IS 6403-1981(Reaffirmed 1997) Relationship between friction angle (ϕ) and SPT 'N'.
3. It is advisable for site inspection of a few trial pits at the start of construction of foundation to revise the safe bearing capacity and depth of foundation based on site conditions accordingly



Dr. LAKSHMI. V

Consulting Geotechnical Engineer

REFERENCES

1. IS 1498 – 1970: Classification and identification of soils for general engineering purposes
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3. IS 2131 – 1981(RA-2016): Method of standard penetration test for soils
4. IS 2720 – 1972: Methods of tests for soils
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6. IS 2950-1982 (Part-1): Code of practice for Design and Construction of Raft foundation
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BORE LOGS, SUB- SOIL PROFILE AND
LABORATORY TEST RESULTS



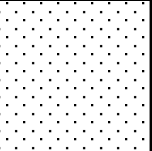
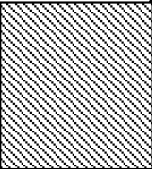
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Project: Geotechnical Investigation for the proposed construction of G+3 Residential building at Talagattapura, Karnataka 560109

Date of Field Investigation: 29.08.2026 **Client:** Mr. Rajkumar Hegade

BH No: 1

Ground water table: Not Encountered

Depth below GL (m)	Soil description	Thickness Of strata (m)	Legend	Details of sampling		SPT N Value	Remarks
				Type	Depth (m)		
0.8	Brownish Silty Sand	0.8		DS	0.8	--	Filled up soil up to 1m
1.5	Brownish Clayey Silty Sand with traces of Gravel	0.7		SPT	1.5	21/42/R N>50	

Borehole was terminated at a depth of 1.5m from the existing ground level

Note :

SPT - Standard penetration test

R - Rebound

UDS - Undisturbed sample

* - Sample not retrieved



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Project: Geotechnical Investigation for the proposed construction of G+3 Residential building at Talagattapura, Bangalore, Karnataka 560109

Date of Field Investigation: 29.08.2026 **Client:** Mr. Rajkumar Hegade

BH No: 2

Ground water table: Not Encountered

Depth below GL (m)	Soil description	Thickness Of strata (m)	Legend	Details of sampling		SPT N Value	Remarks
				Type	Depth (m)		
0.8	Brownish Silty Sand	0.8		DS	0.8	--	Filled up soil up to 0.5m
1.5	Brownish Clayey Silty Sand with traces of gravel	0.7		SPT	1.5	9/18/13 N=31	
3.0	Brownish silty clay with traces of Soft Disintegrated Rock	1.5		SPT	3.0	11/13/14 N=27	
4.0	Soft Disintegrated Rock	1.0		SPT	4.0	23/38/R N>50	

Borehole was terminated at a depth of 4.0m from the existing ground level

Note :

SPT - Standard penetration test

R - Rebound

UDS - Undisturbed sample

* - Sample not retrieved



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Project: Geotechnical Investigation for the proposed construction of G+3 Residential building at Talagattapura, Bangalore, Karnataka 560032

Date of Field Investigation: 29.08.2026 **Client:** Mr. Rajkumar Hegade

BH No: 3

Ground water table: Not Encountered

Depth below GL (m)	Soil description	Thickness Of strata (m)	Legend	Details of sampling		SPT N Value	Remarks
				Type	Depth (m)		
1.5	Brownish Clayey Silty Sand with traces of gravel	1.5		SPT	1.5	12/15/18 N=33	Filled up soil up to 0.5m
2.3	Brownish Clayey silty sand with gravel	0.8		SPT	2.3	27/40/R N>50	

Borehole was terminated at a depth of 2.3m from the existing ground level

Note :

SPT - Standard penetration test

R - Rebound

UDS - Undisturbed sample

* - Sample not retrieved



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APPENDIX - II
LABORATORY TEST RESULTS

Physical Properties:

BORE HOLE	SAMPLE	DEPTH (m)	NMC (%)	GRAIN SIZE DISTRIBUTION			
				GRAVEL (%)	SAND (%)	SILT (%)	CLAY (%)
BH 1	SPT	1.5	10.3	8	60	32	
BH 2	SPT	1.5	5.6	12	58	30	
	SPT	3.0	4.4	6	69	25	
	SPT	4.0	3.8	3	77	20	
BH 3	SPT	1.5	4.6	6	66	28	
	SPT	2.3	3.2	10	70	20	

Atterberg Limits

BORE HOLE	SAMPLE	DEPTH (m)	Atterberg Limits	
			LL (%)	PL (%)
BH 1	SPT	1.5	28	16
BH 2	SPT	1.5	24	12
	SPT	3.0	20	8
	SPT	4.0	18	NP
BH 3	SPT	1.5	22	10
	SPT	2.3	20	10

Shear Parameters

BORE HOLE	SAMPLE	DEPTH (m)	Dry Density, kN/m ²	Shear Parameters	
				Cohesion, kN/m ²	Angle of Internal friction, degrees
BH 1	SPT	1.5	18.5	12.2	28
BH 2	SPT	1.5	18.5	13.6	26
	SPT	3.0	19.2	10.2	26
	SPT	4.0	19.9	2.6	28
BH 3	SPT	1.5	19.0	11.8	28
	SPT	2.3	19.2	10.8	28

APPENDIX-III

TYPICAL CALCULATION OF ALLOWABLE BEARING PRESSURE

Ref: I.S. 2131 – 1963, I.S. 6403-1981, I.S. 8009 (PART – I)-1976

Sample Identification: BH.No.-1

Soil Parameters:

At a depth of 1.5m and B=1m, $c = 12.2\text{kPa}$, $\phi = 28^\circ$, and $\gamma = 1.85\text{ t/m}^3$
Considering local shear failure, $\phi_m = \tan^{-1}(0.67 \tan\phi) = 19.6^\circ$, $W'=1$

SHEAR FAILURE CONDITION

Bearing capacity factors:

$$\begin{aligned}N_q &= e^{\pi \tan \phi} \tan^2 (45 + \phi_m/2) \\&= e^{\pi \tan 28^\circ} \tan^2 (45 + 19.6/2) \\&= 6.2\end{aligned}$$

$$\begin{aligned}N_c &= (N_q - 1) \cot \phi_m \\&= (6.2-1) \cot 19.6^\circ \\&= 14.5\end{aligned}$$

$$\begin{aligned}N_\gamma &= 2 (N_q+1) \tan \phi_m \\&= 5.8\end{aligned}$$

Depth factors:

$$\begin{aligned}d_c &= 1+0.2 (D_f /B) \tan (45 + \phi/2) \\&= 1+ 0.2 (1.5/2) \tan (45 +28/2) \\&= 1.49\end{aligned}$$

$$\begin{aligned}d_q &= d_\gamma = 1 + 0.1 ((D_f /B) \tan (45 + \phi/2) \\&= 1+ 0.1 (1.5/2) \tan (45 +28/2) \\&= 1.25\end{aligned}$$

Shape factors: $S_c = 1.3$, $S_q = 1.2$, $S_\gamma = 0.8$

Substituting the above factors in,

$$Q_f = cN_cS_c d_c + \gamma D(N_q- 1) S_q d_q + 0.5\gamma B N_\gamma S_\gamma d_\gamma W'$$

$$Q_f = 498.53\text{ kN/m}^2 = 49.85\text{ t/m}^2$$

Using a factor of safety of 3.0, $Q_a = 166.18\text{ kN/m}^2 = 16.62\text{ t/m}^2$

SETTLEMENT CRITERIA

FROM SPT N – VALUE

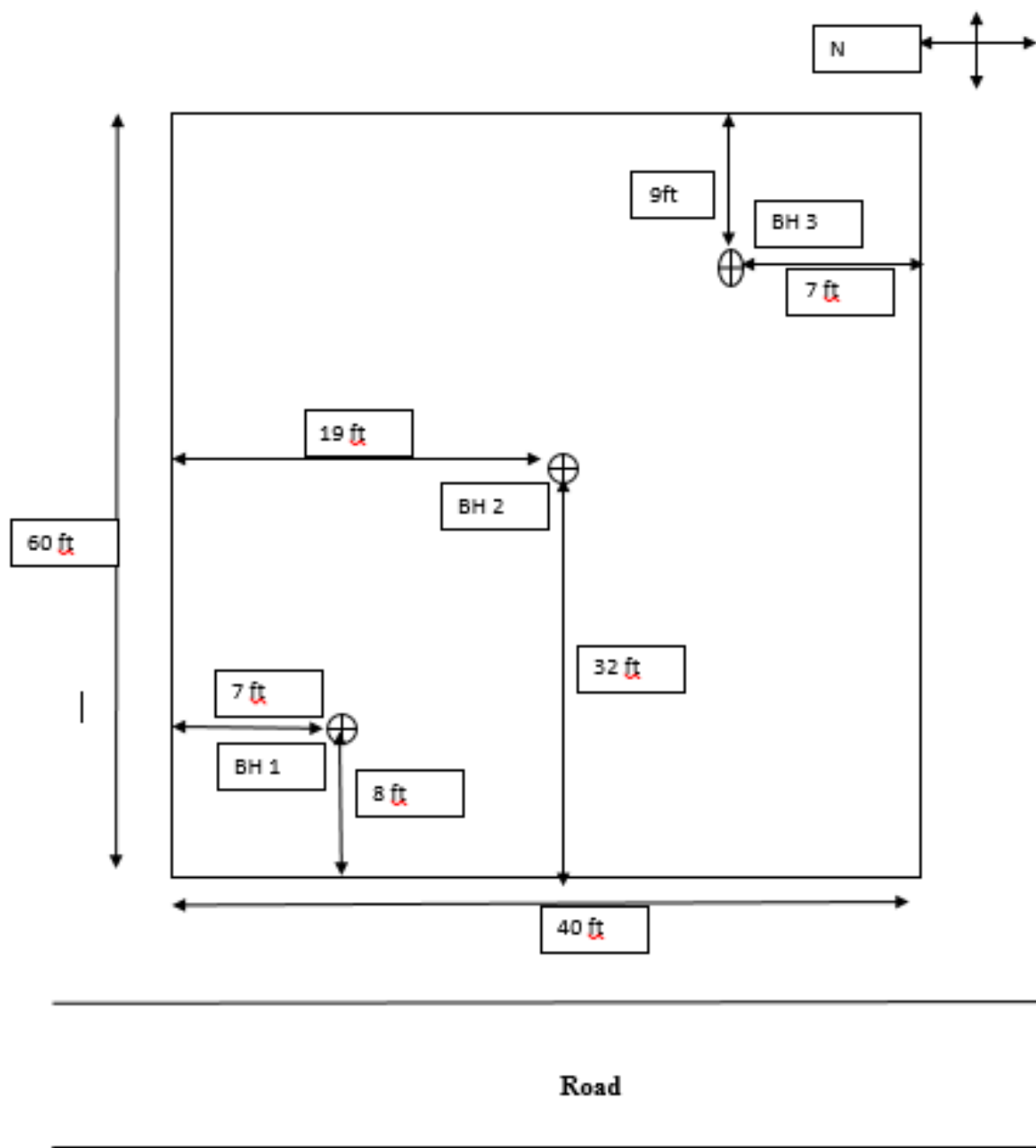
For a N – value of 50 at a depth of 1.5m, settlement for a pressure of 1 kg / cm^2 for B = 1.0m is 5 mm.

$$\begin{aligned}\text{For an allowable settlement of } 25\text{mm, } Q_a &= (25) \div 5 \\&= 5\text{ kg / cm}^2 \text{ or } 50\text{ tons /m}^2\end{aligned}$$

The lower of the above two is taken as the allowable bearing pressure.

Note: The allowable bearing pressure at 1.5m depth is recommended as 160kN/m^2 OR 16t/m^2 as a conservative estimate

LOCATION PLAN OF BORE HOLE







Standard Penetration Test Progress and Soil sample inside the split spoon sampler (BH- 1, 2 & 3)